

Computational Fluid Dynamics for Aerospace Applications

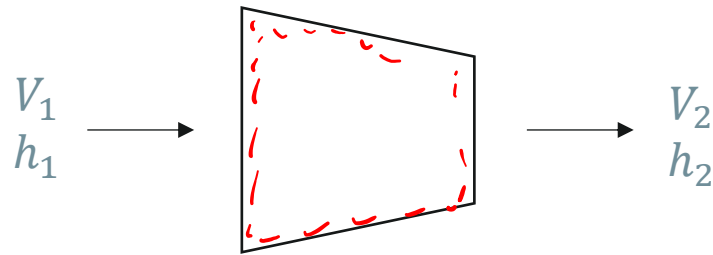
Lecture 3 Governing Equations 2

Governing Equations

- Energy
- Conservation
- Equation of State
- Inviscid Flow
- Incompressible Flow

Energy

- We previously derived mass and momentum equations
- Is that enough for us to describe a general, compressible flow?
- Example: Thermodynamic analysis of flow through a nozzle



First Law, steady, neglect PE, constant C_p

$$h_1 + \frac{1}{2}V_1^2 = h_2 + \frac{1}{2}V_2^2$$

$$\frac{1}{2}(V_2^2 - V_1^2) = h_1 - h_2$$

$$\frac{1}{2}(V_2^2 - V_1^2) = C_p(T_1 - T_2)$$

- Energy (or enthalpy, or temperature, etc.) in a compressible flow is linked to velocity → we need to derive an energy equation

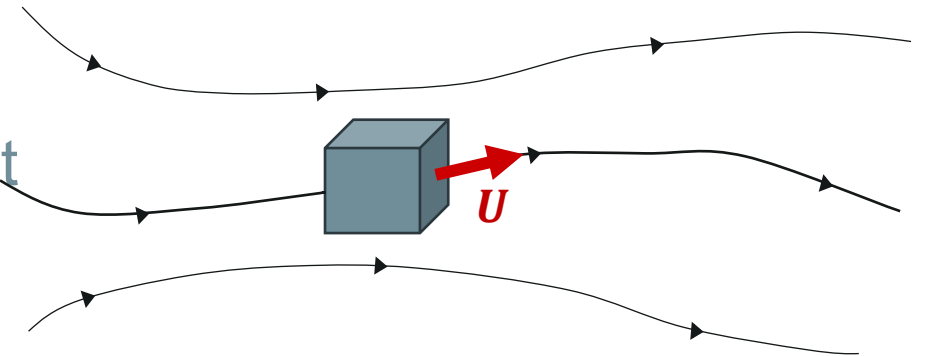
$$C_p \approx 1000 \frac{\text{J}}{\text{kg K}}$$

Energy

- Energy equation is derived from first principles
 - Energy is conserved
 - Analyze an infinitesimally small element moving with the flow
- First Law of Thermodynamics
 - Because we are tracking a moving mass of fluid, this is a closed system

$$\bullet \frac{D(\text{Energy})}{Dt} = \dot{Q}_{in} + \dot{W}_{in}$$

- (A) • 1) Rate of change of energy inside element
- (B) • 2) Net flux of heat into element
- (C) • 3) Rate of surface/body work on element



power = force x velocity

Energy

- Rate of surface/body work on element

- $\dot{W} = \dot{W}_{body} + \dot{W}_{surf} = \sum \mathbf{F} \cdot \mathbf{V}$

- $\dot{W}_{body} = \sum F_{b,x} u + \sum F_{b,y} v + \sum F_{b,z} w$

$$\dot{W}_b = (\text{mass}) \left(\frac{\text{force}}{\text{mass}} \right) (\text{velocity})$$

$$\dot{W}_b = (\rho dx dy dz) (\vec{f}) \cdot (\vec{V})$$

$$\dot{W}_{body} = \rho dx dy dz \mathbf{f} \cdot \mathbf{V}$$

Energy

- (C) • Rate of surface/body work on element

$$\dot{W}_{surf} = \sum F_{s,x} u + \sum F_{s,y} v + \sum F_{s,z} w$$

X-direction: (Surface Power)

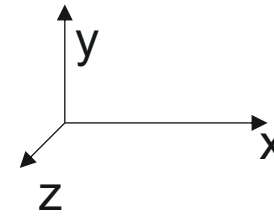
Pressure: $u p dy dz - \left(u p + \frac{\partial(u p)}{\partial x} dx \right) dy dz$

$$= - \frac{\partial(u p)}{\partial x} dx dy dz$$

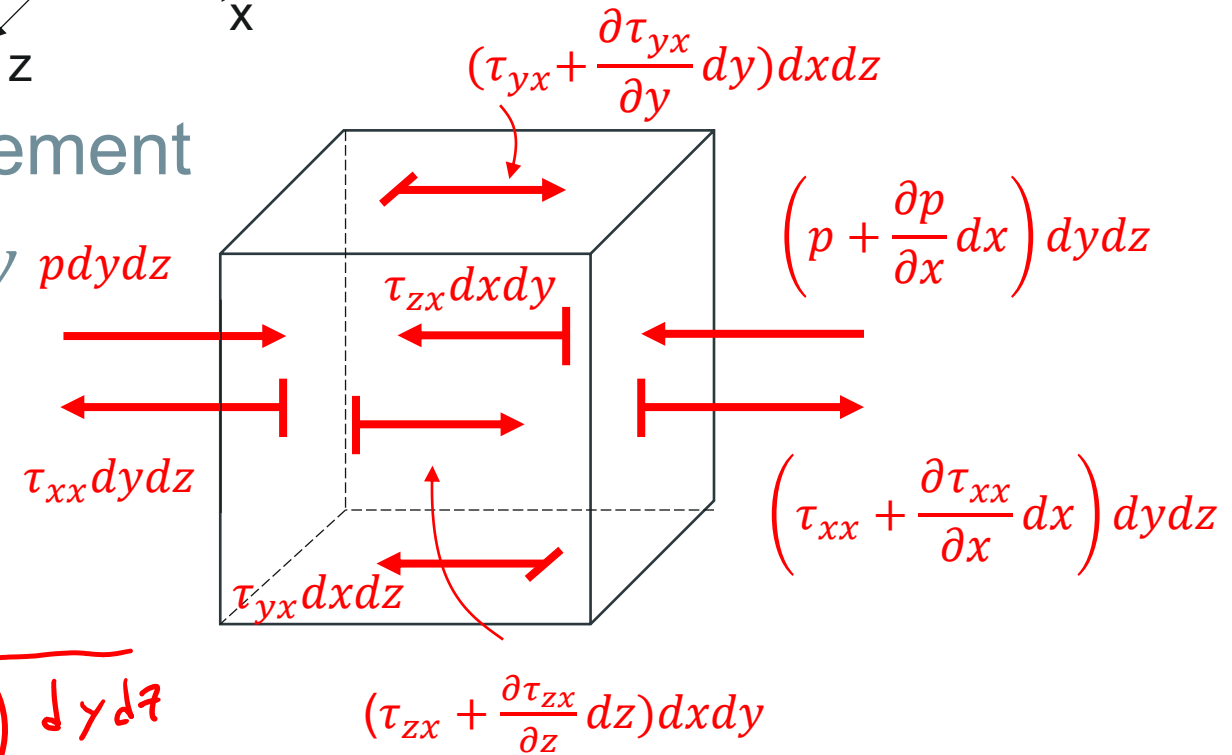
τ_{xx} term: $u \tau_{xx} dy dz - \left(u \tau_{xx} + \frac{\partial(u \tau_{xx})}{\partial x} dx \right) dy dz$

$$= - \frac{\partial(u \tau_{xx})}{\partial x} dx dy dz$$

τ_{yx} : $- \frac{\partial(u \tau_{yx})}{\partial x} dx dy dz$ | τ_{zx} : $- \frac{\partial(u \tau_{zx})}{\partial x} dx dy dz$



Surface Forces



Sum

Energy

Sum x, y, z
because energy is
a scalar

(C)

- Rate of surface/body work on element

- $\dot{W} = \dot{W}_{body} + \dot{W}_{surf} = \sum \mathbf{F} \cdot \mathbf{V}$

- $\dot{W}_{surf} = \sum F_{s,x}u + \sum F_{s,y}v + \sum F_{s,z}w$

Last slide
↓

*

$$\dot{W}_{surf,x} = \left(\left[-\frac{\partial up}{\partial x} + \frac{\partial u\tau_{xx}}{\partial x} + \frac{\partial u\tau_{yx}}{\partial y} + \frac{\partial u\tau_{zx}}{\partial z} \right] \right) dx dy dz$$

$$\dot{W}_{surf,y} = \left(\left[-\frac{\partial vp}{\partial y} + \frac{\partial v\tau_{xy}}{\partial x} + \frac{\partial v\tau_{yx}}{\partial y} + \frac{\partial v\tau_{zy}}{\partial z} \right] \right) dx dy dz$$

$$\dot{W}_{surf,z} = \left(\left[-\frac{\partial wp}{\partial z} + \frac{\partial w\tau_{xz}}{\partial x} + \frac{\partial w\tau_{yz}}{\partial y} + \frac{\partial w\tau_{zz}}{\partial z} \right] \right) dx dy dz$$

$$\begin{aligned} \dot{W} &= \left(\left[-\left(\frac{\partial up}{\partial x} + \frac{\partial vp}{\partial y} + \frac{\partial wp}{\partial z} \right) \right. \right. \\ &\quad + \frac{\partial u\tau_{xx}}{\partial x} + \frac{\partial u\tau_{yx}}{\partial y} + \frac{\partial u\tau_{zx}}{\partial z} \\ &\quad + \frac{\partial v\tau_{xy}}{\partial x} + \frac{\partial v\tau_{yx}}{\partial y} + \frac{\partial v\tau_{zy}}{\partial z} \\ &\quad \left. \left. + \frac{\partial w\tau_{xz}}{\partial x} + \frac{\partial w\tau_{yz}}{\partial y} + \frac{\partial w\tau_{zz}}{\partial z} \right] \right. \\ &\quad \left. + \rho \mathbf{f} \cdot \mathbf{V} \right) dx dy dz \end{aligned}$$

Energy

- ③ • Net flux of heat into element
 - Volumetric heating (e.g., absorption or emission of radiation)

$$\dot{Q}_v = (\text{mass}) \left(\frac{\text{heat}}{\text{mass}} \right)$$

↘ or
Chemical
reaction

$$\dot{Q}_v = (\rho dx dy dz) \dot{q}$$

$$\dot{Q}_v = \rho \dot{q} dx dy dz$$

Surface Heating

②

Energy

- Net flux of heat into element
- Surface heating (e.g., due to temperature gradients like in conduction)

x-direction:

$$\left[\dot{q}_x - \left(\dot{q}_x + \frac{\partial \dot{q}_x}{\partial x} dx \right) \right] dy dz$$

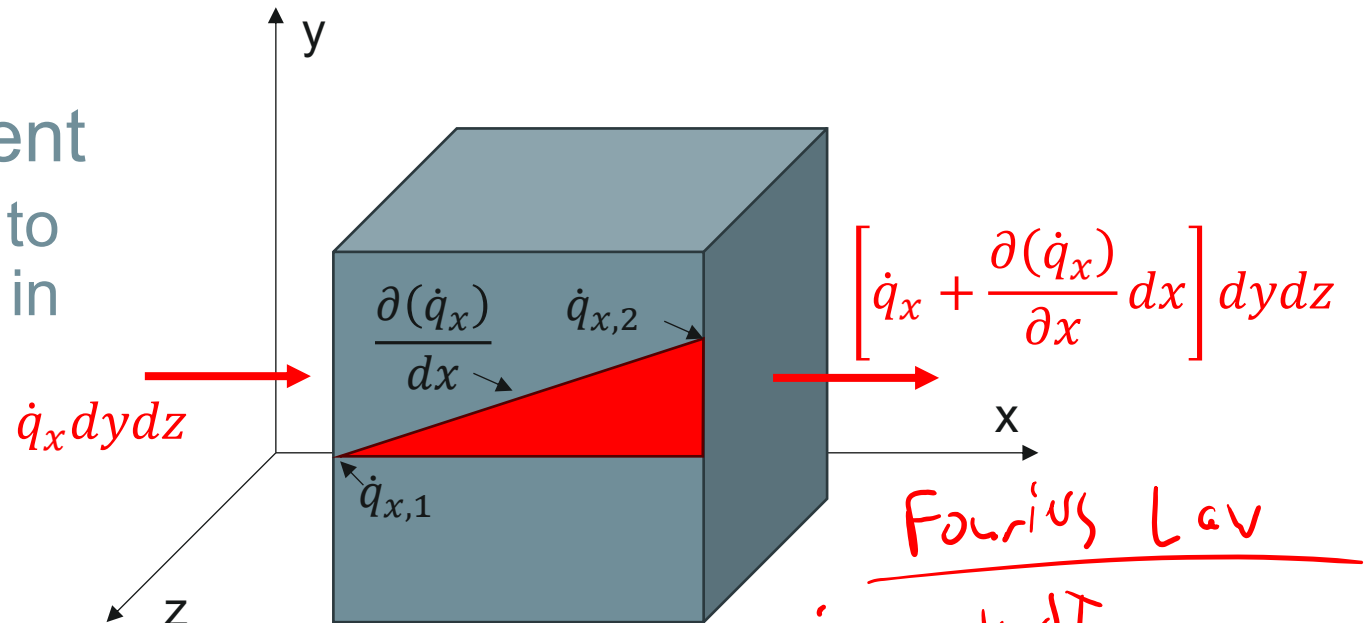
$$= - \frac{\partial \dot{q}_x}{\partial x} dx dy dz$$

Total in all directions
+ Volumetric

$$\left[\rho \dot{q} - \frac{\partial \dot{q}_x}{\partial x} - \frac{\partial \dot{q}_y}{\partial y} - \frac{\partial \dot{q}_z}{\partial z} \right] dx dy dz$$

*

$$\dot{Q}_s = \left[\rho \dot{q} + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \right] dx dy dz$$



$$\begin{aligned} \dot{q}_x &= -k \frac{\partial T}{\partial x} \\ \dot{q}_y &= -k \frac{\partial T}{\partial y} \\ \dot{q}_z &= -k \frac{\partial T}{\partial z} \end{aligned}$$

Energy

(A)

- Rate of change of energy inside element
- $\frac{D(Energy)}{Dt} \rightarrow$ but what energy exactly is this?
- In thermodynamics, we consider total energy to be the sum of:
 - ✓ • Kinetic energy (energy stored in bulk fluid motion)
 - ✗ • Potential energy (ability to do work through gravity)
 - Internal energy (microscopic forms of energy, i.e., in molecules/atoms)
 - ✓ • Sensible energy (molecular translation and rotation, relates to temperature)
 - ✗ • Latent energy (energy between molecules, relates to phase change)
 - ✗ • Chemical energy (energy stored in bonds within a molecule)
 - ✗ • Nuclear energy (energy stored within nucleus of atoms)

Energy

- Rate of change of energy inside element
- $\frac{D(\text{Energy})}{Dt} \rightarrow$ but what energy exactly is this?
- For our applications, we will consider:
 - Kinetic energy $V^2/2$
 - Internal energy e (specifically, sensible energy)

$$\rho \frac{D}{Dt} \left(e + \frac{V^2}{2} \right) dx dy dz$$

- If you want to model a system that includes large changes in other types of energy, those types of energies need to be considered
 - E.g., A simulation of combustion must include chemical internal energy
 - E.g., A simulation of water boiling must include latent internal energy

Energy

- Combining everything with the first law, we find

$$\begin{aligned} & \rho \frac{D}{Dt} \left(e + \frac{V^2}{2} \right) \\ &= \rho \dot{q} + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \\ & - \left(\frac{\partial up}{\partial x} + \frac{\partial vp}{\partial y} + \frac{\partial wp}{\partial z} \right) + \frac{\partial u \tau_{xx}}{\partial x} + \frac{\partial u \tau_{yx}}{\partial y} + \frac{\partial u \tau_{zx}}{\partial z} + \frac{\partial v \tau_{xy}}{\partial x} \\ & + \frac{\partial v \tau_{yx}}{\partial y} + \frac{\partial v \tau_{zy}}{\partial z} + \frac{\partial w \tau_{xz}}{\partial x} + \frac{\partial w \tau_{yz}}{\partial y} + \frac{\partial w \tau_{zz}}{\partial z} + \rho \mathbf{f} \cdot \mathbf{V} \end{aligned}$$

Energy

- ① convert $\partial/\partial t \rightarrow d/dt$
- ② plug in for τ_{ij}
- ③ derive DV^2/Dt

- There are several forms of the equation through manipulation
 - For example, regular derivative of just internal energy plugging in for τ 's
 - Note: This involves deriving a governing equation for $\frac{DV^2}{Dt}$ starting from momentum and subtracting it out to isolate the internal energy
 - Note the lack of a body force, it won't affect the internal energy, only kinetic energy

$$\begin{aligned}
 & \frac{\partial(\rho e)}{\partial t} + \frac{\partial(\rho eu)}{\partial x} + \frac{\partial(\rho ev)}{\partial y} + \frac{\partial(\rho ew)}{\partial z} \\
 &= \rho \dot{q} + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) - p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) - \frac{2}{3} \mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)^2 \\
 &+ \mu \left[2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + 2 \left(\frac{\partial w}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)^2 \right]
 \end{aligned}$$

Energy

- What do these various terms mean?

rate of change
of the local
internal energy

advection of internal energy

diffusion of energy
due to thermal
gradients

$p \nabla \cdot \vec{V}$ pressure

work
associated
with volume
changes

Source term
(heat)

$$\begin{aligned} & \frac{\partial(\rho e)}{\partial t} + \frac{\partial(\rho eu)}{\partial x} + \frac{\partial(\rho ev)}{\partial y} + \frac{\partial(\rho ew)}{\partial z} \\ &= \boxed{\rho \dot{q}} + \underbrace{\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)}_{\text{diffusion of energy due to thermal gradients}} - \underbrace{p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)}_{p \nabla \cdot \vec{V} \text{ pressure work associated with volume changes}} - \underbrace{\frac{2}{3} \mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)^2}_{\text{viscous heating} \rightarrow \text{irreversible}} \\ &+ \mu \left[2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + 2 \left(\frac{\partial w}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)^2 \right] \end{aligned}$$

Viscous heating $\rightarrow$ irreversible

Conservation

- We've derived equations for mass (1), momentum (3), and energy (1)
- Each equation was shown in multiple form, but which to use?
 - All forms of these equations are correct and can give correct answers
 - Some forms are easier to use and have mathematical benefits
- The most important distinction is between conservative and nonconservative forms of the equations

Conservation

- Conservative form (maintains flux *inside* divergence)
 - For example, a conservative form of momentum
 - $\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{V}) = RHS$
- Nonconservative form (expands the flux *outside* divergence)
 - For example, a nonconservative form of momentum
 - $\rho \frac{\partial(u)}{\partial t} + \rho \mathbf{V} \cdot \nabla u = RHS$
- Mathematically, both are identical until we start approximating the various terms
- Conservative form ensures quantity is conserved (e.g., mass is conserved) up to tolerances, nonconservative does not

Is this theoretically solvable?

- Assume we can evaluate $f, \dot{q}, \mu, k$
- Mass equation (1): $f(\rho, u, v, w)$
- Momentum equations (3): $f(\rho, u, v, w, p)$
- Energy equation (1): $f(\rho, u, v, w, p, e, T)$
- Five equations, seven unknowns
 - How do we add equations without adding unknowns?

Equation of State

- Relate temperature, pressure, and density through EOS
 - We will generally assume ideal gas
 - $p = \rho RT$
- Relate variables to sensible internal energy
 - $e = e(T, p)$ for a general fluid
 - $e = e(T)$ for an ideal gas
 - We will generally assume constant specific heats for simplicity, but recall this assumption becomes bad as temperature changes become large
 - $e = c_v T$
- We now have seven equations and seven unknowns

Inviscid Flow

- We derived very general equations for high-speed flows
 - What if our flow doesn't exhibit all physics described by our equations?
 - For example, a high Reynolds number flow has minimal viscous effects and thus these can *sometimes* be ignored without affecting the solution

$$Re = \frac{\rho u L}{\mu} \gg 1 \rightarrow \mu \ll \rho u L$$

- In this case, we can approximate that viscosity (and potentially other diffusive effects are negligible)
 - $\mu \approx 0$
 - $k \approx 0$
 - $D \approx 0$

Inviscid Flow

- This gives us “Euler’s Equations”

Mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

$$p = \rho RT$$

$$e = c_v T$$

Momentum X

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho uu)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} = -\frac{\partial p}{\partial x} + \rho f_x$$

Momentum Y

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho vu)}{\partial x} + \frac{\partial(\rho vv)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} = -\frac{\partial p}{\partial y} + \rho f_y$$

Momentum Z

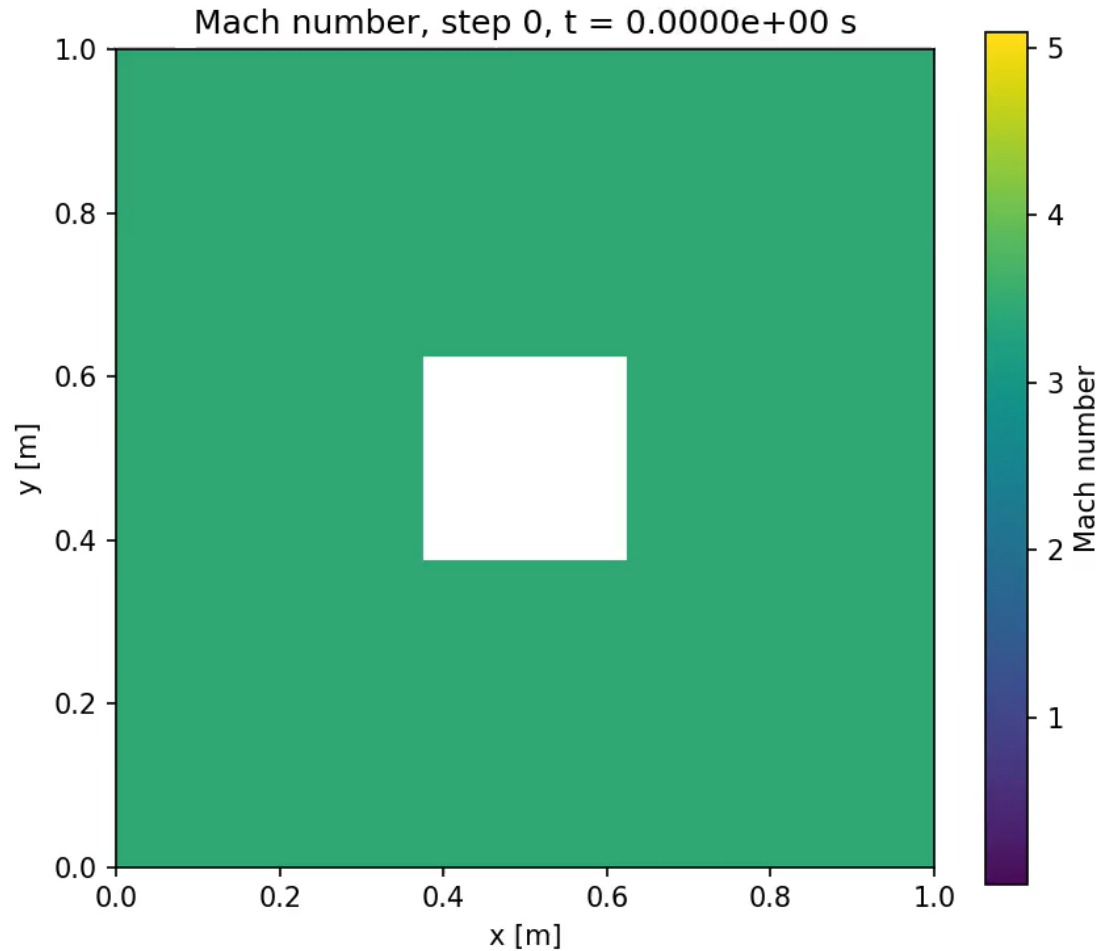
$$\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho wu)}{\partial x} + \frac{\partial(\rho wv)}{\partial y} + \frac{\partial(\rho ww)}{\partial z} = -\frac{\partial p}{\partial z} + \rho f_z$$

Energy

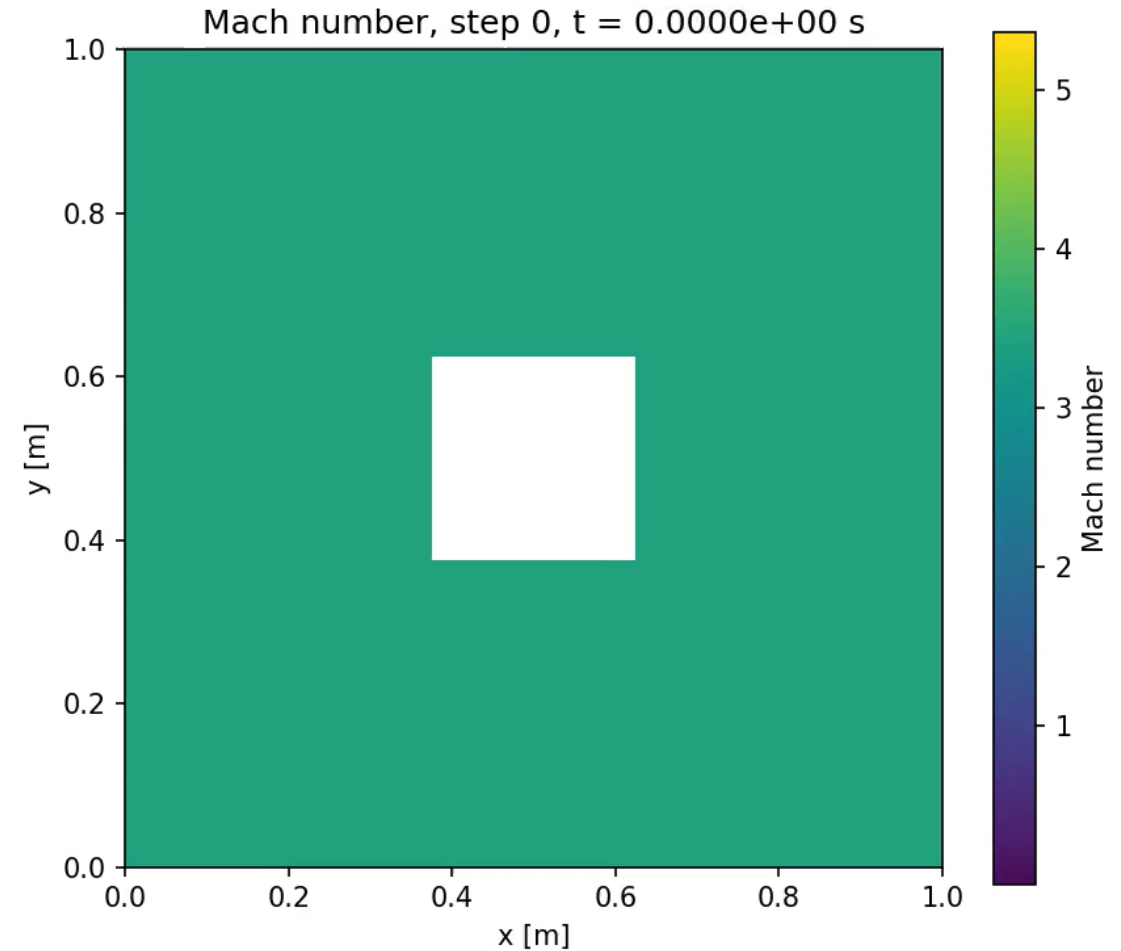
$$\frac{\partial(\rho e)}{\partial t} + \frac{\partial(\rho eu)}{\partial x} + \frac{\partial(\rho ev)}{\partial y} + \frac{\partial(\rho ew)}{\partial z} = \rho \dot{q} - p \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)$$

Inviscid Flow

Full equations



Euler equations



Incompressible Flow

- At slow speeds ($M < 0.3$), fluids are largely incompressible
 - This leads to simpler equations, but a more complicated solution process
 - We will not focus on incompressible flows in this course
- Material derivatives of density go to zero
 - $\frac{D\rho}{Dt} \rightarrow 0$

Mass

$$\nabla \cdot \mathbf{V} = 0$$

Momentum

$$\rho \frac{D\mathbf{V}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{V} + \rho \mathbf{f}$$

- Energy equation gone, coupling between velocity and internal energy negligible, i.e., change in flow won't change temperature

Looking Forward

- We now know exactly which equations need to be solved
 - No analytical solution exists for these equations
 - How do we solve them?
-
- Next lecture, we will learn a little bit about different types of PDEs and learn how we can start approximating their solutions